

5 Queries (MongoDB):

1. Presio altuko alertak (> 5,5 bar) gune zehatzetan

```
db.data.aggregate([
  {
    $match: {
      pressure_bar: { $gt: 5.5 }
    }
  },
  {
    $group: {
      _id: {
        site: "$siteId",
        sensor: "$sensorId"
      },
      alertas: { $sum: 1 }
    }
  }
])
```

```
< {
  _id: {
    site: 'PLANT-EIBAR-02',
    sensor: 'FLOW-METER-78'
  },
  alertas: 3446
}
{
  _id: {
    site: 'PLANT-EIBAR-02',
    sensor: 'FLOW-METER-77'
  },
  alertas: 3511
}
```

2. pH balio arriskutsuak bakarrik bilatu

```
db.data.aggregate([
  {
    $match: {
```

```
$or: [  
  { phLevel: { $lt: 7.0 } },  
  { phLevel: { $gt: 7.5 } }  
]  
}  
,  
{  
  $project: {  
    _id: 0,  
    timestamp: 1,  
    siteId: 1,  
    phLevel: 1,  
    alerta: { $literal: "CHECK_CHEMICALS" }  
  }  
}
```

])

```
{
  timestamp: 2026-02-12T10:50:43.084Z,
  siteId: 'PLANT-EIBAR-02',
  phLevel: 7.72,
  alerta: 'CHECK_CHEMICALS'
}
{
  timestamp: 2026-02-12T10:51:07.109Z,
  siteId: 'PLANT-EIBAR-01',
  phLevel: 6.87,
  alerta: 'CHECK_CHEMICALS'
}
{
  timestamp: 2026-02-12T10:51:07.109Z,
  siteId: 'PLANT-EIBAR-02',
  phLevel: 7.65,
  alerta: 'CHECK_CHEMICALS'
}
{
  timestamp: 2026-02-12T10:51:13.111Z,
  siteId: 'PLANT-EIBAR-02',
  phLevel: 7.63,
  alerta: 'CHECK_CHEMICALS'
}
{
  timestamp: 2026-02-19T09:38:32.470Z,
  siteId: 'PLANT-EIBAR-01',
  phLevel: 6.87,
  alerta: 'CHECK_CHEMICALS'
}
```

3. Erregistratutako 5 fluxu handienak

```
db.data.aggregate([
  { $sort: { flowRate_Lpm: -1 } },
  { $limit: 5 },
  {
```

```

$project: {
  _id: 0,
  siteId: 1,
  sensorId: 1,
  Max_flowRate: "$flowRate_Lpm",
}
}
])

```

```

{
  siteId: 'PLANT-EIBAR-02',
  sensorId: 'FLOW-METER-77',
  Max_flowRate: 160
}
{
  siteId: 'PLANT-EIBAR-02',
  sensorId: 'FLOW-METER-77',
  Max_flowRate: 160
}
{
  siteId: 'PLANT-EIBAR-02',
  sensorId: 'FLOW-METER-77',
  Max_flowRate: 160
}
{
  siteId: 'PLANT-EIBAR-02',
  sensorId: 'FLOW-METER-78',
  Max_flowRate: 160
}
{
  siteId: 'PLANT-EIBAR-02',
  sensorId: 'FLOW-METER-77',
  Max_flowRate: 160
}
}

```

4.PhLevel eta Pressure Bar txikiena duten 5 erregistro

```

db.data.aggregate([
  {
    $sort: {

```

```
    phLevel: 1,  
    pressure_bar: 1  
  }  
},  
{ $limit: 5 },  
{  
  $project: {  
    _id: 0,  
    siteId: 1,  
    sensorId: 1,  
    Min_phLevel: "$phLevel",  
    Min_pressurebar: "$pressure_bar"  
  }  
}  
}  
])
```

```

{
  siteId: 'PLANT-EIBAR-01',
  sensorId: 'FLOW-METER-78',
  Min_phLevel: 6.8,
  Min_pressurebar: 2.39
}
{
  siteId: 'PLANT-EIBAR-01',
  sensorId: 'FLOW-METER-78',
  Min_phLevel: 6.8,
  Min_pressurebar: 4.51
}
{
  siteId: 'PLANT-EIBAR-01',
  sensorId: 'FLOW-METER-78',
  Min_phLevel: 6.8,
  Min_pressurebar: 4.52
}
{
  siteId: 'PLANT-EIBAR-01',
  sensorId: 'FLOW-METER-78',
  Min_phLevel: 6.8,
  Min_pressurebar: 4.52
}
{
  siteId: 'PLANT-EIBAR-01',
  sensorId: 'FLOW-METER-78',
  Min_phLevel: 6.8,
  Min_pressurebar: 4.52
}

```

5.Sensor bakoitzean egon diren alerta kopurua

```

db.data.aggregate([
  { $match: { status: "LEAK_DETECTED" } },
  {
    $group: {
      _id: "$sensorId",
      total_fugas: { $sum: 1 }
    }
  }
])

```

```
}  
}  
])
```

```
{  
  _id: 'FLOW-METER-77',  
  total_fugas: 395  
}  
{  
  _id: 'FLOW-METER-78',  
  total_fugas: 413  
}
```